

Math 135

Please give me feedback! I am at the first row.

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§1 Admin

§1.1 General Information

§1.1.1 Contact

For mathematics questions: abeltaos@uwaterloo.ca

For course logistics: math135@uwaterloo.ca

§1.1.2 Teaching Style

The LEARN website contains several useful resources. In particular:

- (i) The icons at the top of the page provide quick access to important resources, including the course schedule.
- (ii) Crowdmark is used for submitting assignments.

§1.1.3 Graded Assignments

There are three assignments each week.

§1.1.4 Late Submission Policy

Late submissions are accepted with a penalty of 1% per minute for the entire assignment.

§1.1.5 Office Hours

Monday and Wednesday 2:00 to 3:30 in MC 6503 or by appointment (email the instructor to arrange).

§2 Introduction to the Language of Mathematics

§2.1 Introduction to Proof

Checking a few examples may suggest that a statement is true. A proof explains why it holds for every object covered by the statement.

Definition 2.1.1 (Proof)

A proof is a logically valid argument that establishes the truth of a mathematical statement using definitions, axioms, and previously proved results.

Proposition 2.1.2

For every positive integer n , the number $n^2 + 1$ is not a perfect square.

Proof. Let n be a positive integer. The consecutive perfect squares surrounding $n^2 + 1$ are n^2 and $(n + 1)^2$. Indeed,

$$n^2 < n^2 + 1 < n^2 + 2n + 1 = (n + 1)^2,$$

where the second inequality follows from $n > 0$.

Therefore, $n^2 + 1$ lies strictly between two consecutive perfect squares, so it cannot itself be a perfect square. *Q.E.D.*

In practice

- **Ruling out a perfect square.** Locate the number strictly between two consecutive perfect squares. Check both inequalities and identify where the hypothesis on the number is used.

§2.2 Sets

Definition 2.2.1 (Set)

A **set** is a well-defined, unordered collection of distinct objects. Each object that appears in this collection is called an **element** of the set.

For a finite set, we can denote it by listing its elements, separated by commas, surrounding the entire list by braces $\{ \}$.

Example 2.2.2

The set containing the elements 2, 7, and -3 can be denoted by

$$\{7, -3, 2\}.$$

The order of listing is irrelevant, so $\{2, 7, -3\}$ is the same set.

Example 2.2.3

Here are some examples:

- $\{2, 4, 6, 8\}$
- $\{1, 2, \{1, 2, 3\}\}$
- $\{a, b, c, d\}$
- $\mathbb{N} = \{1, 2, 3, \dots\}$
- $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$
- $\mathbb{Q} = \{\frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0\}$

An element may itself be a set: $\{1, 2, \{1, 2, 3\}\}$ has exactly three elements, namely 1, 2, and $\{1, 2, 3\}$.

Remark. Infinite sets can have different sizes: $|\mathbb{N}| = \aleph_0$ while $|\mathbb{R}| = 2^{\aleph_0}$, and $2^{\aleph_0} > \aleph_0$. Whether $2^{\aleph_0}$ equals the next size $\aleph_1$ is the *continuum hypothesis*, which can be neither proved nor disproved from the standard axioms, so $|\mathbb{R}| = \aleph_1$ should not be written as a fact.

Definition 2.2.4

The **parity** of an integer is **even** if the integer is a multiple of 2, and **odd** otherwise.

§2.3 Statement and Negation

Definition 2.3.1 (Statement)

A **statement** is a sentence that has a definite state of being either true or false.

Definition 2.3.2 (Negation)

Suppose that A is a statement. Then the negation of A , denoted by $\neg A$, is the statement asserting the opposite truth value to A . That is, $\neg A$ is false when A is true, and $\neg A$ is true when A is false.

§2.4 Quantifiers and Quantified Statements

Quantifiers specify how many elements of a domain must satisfy a condition. When quantifiers are nested, their order determines which choices may depend on earlier ones.

§2.4.1 Universal Quantifier

$\forall$ means “for all”, and it is called the universal quantifier. The statement $\forall x \in S, P(x)$ is true when $P(x)$ is true for *every* element x of S .

Example 2.4.1

“Every integer $n \geq 5$ satisfies $2^n > n^2$ ” can be written as

$$\forall n \in \{n \in \mathbb{Z} : n \geq 5\}, 2^n > n^2.$$

Checking $n = 5, 6, 7$ is not a proof: a universal statement must hold for every n in the domain. (The domain matters: the statement fails at $n = 3$, since $8 < 9$.)

The negation of a universally quantified statement is

$$\neg (\forall x \in S, P(x)) \equiv (\exists x \in S, \neg P(x)).$$

So a single counterexample is enough to show that a universal statement is false.

§2.4.2 Existential Quantifier

$\exists$ means “there exists”, and it is called the existential quantifier. The statement $\exists x \in S, P(x)$ is true when $P(x)$ is true for *at least one* $x \in S$.

Example 2.4.2

$$\exists a \in \mathbb{Z}, a^2 + 29a + 209 \leq 0.$$

This is true: one witness is enough, and $a = -14$ gives $196 - 406 + 209 = -1 \leq 0$.

The negation of an existentially quantified statement (such as $\exists x \in S, P(x)$) is

$$\neg (\exists x \in S, P(x)) \equiv (\forall x \in S, \neg P(x)).$$

Remark. A quantified statement contains four parts:

- variable;
- quantifier;
- domain;
- an open sentence involving the variable (that is either true or false whenever a value of the variable is chosen from the domain).

§2.4.3 Nested Quantifiers

Consider the statement

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R}, x > y.$$

This statement is true: given any $x \in \mathbb{R}$, we may choose $y = x - 1$, and then $x > y$. Note that y is allowed to depend on x .

Remark. Reversing the quantifiers changes the statement: $\exists y \in \mathbb{R}, \forall x \in \mathbb{R}, x > y$. It says there is a single real number y which is smaller than every real number, including y itself. This is false: taking $x = y$ gives $y > y$.

To negate a nested quantified statement, negate one quantifier at a time from the outside in: each $\forall$ becomes $\exists$, each $\exists$ becomes $\forall$, and the open sentence is negated. For example,

$$\begin{aligned} \neg (\forall x \in S, \exists y \in T, P(x, y)) \\ \equiv \exists x \in S, \forall y \in T, \neg P(x, y). \end{aligned}$$

In practice

- **Reading quantifiers.** State the domain and read from left to right. In $\forall x \in S, \exists y \in T, P(x, y)$, the choice of y may depend on x ; reversing the quantifiers requires one y that works for every x .
- **Negating a statement.** Keep the domains and quantifier order, swap each $\forall$ with $\exists$, and negate the final condition. One counterexample refutes a universal statement; refuting an existential statement requires ruling out every possible witness.

§2.5 Summation and Product Notation

Writing $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$ is manageable, but writing out every term through 1000 is not. Summation notation specifies both the terms and their range, instead of relying on dots in $1 + 2 + \cdots + 1000$.

Definition 2.5.1 (Summation notation)

For integers $a \leq b$ and terms $t_a, \dots, t_b$, write

$$\sum_{i=a}^b t_i = t_a + t_{a+1} + \cdots + t_b.$$

The index i runs through every integer from a to b .

Example 2.5.2

$$1 + 2 + \cdots + 100 = \sum_{i=1}^{100} i, \quad 1 + 8 + 27 + 64 + 125 = \sum_{i=1}^5 i^3.$$

In the second sum, the term indexed by i is i^3 .

§2.5.1 Properties of Summations

For terms x_i, y_i , a constant c , and integer bounds $a \leq b$ and $m \leq n$, the following identities let us rewrite sums without changing their terms:

(i) Multiplying by a constant:

$$c \sum_{i=a}^b x_i = \sum_{i=a}^b cx_i.$$

(ii) Adding or subtracting sums with the same bounds:

$$\sum_{i=a}^b x_i \pm \sum_{i=a}^b y_i = \sum_{i=a}^b (x_i \pm y_i).$$

(iii) Shifting the index by an integer r :

$$\sum_{i=m}^n x_i = \sum_{j=m+r}^{n+r} x_{j-r}.$$

The new index $j = i + r$ names the same terms.

(iv) Splitting a sum at n , where $2 \leq n \leq k$:

$$\sum_{i=1}^{n-1} x_i + \sum_{i=n}^k x_i = \sum_{i=1}^k x_i.$$

Example 2.5.3

Shifting by $r = 1$ and splitting off the last term,

$$\sum_{i=0}^9 (i+1) = \sum_{j=1}^{10} j = \sum_{j=1}^9 j + 10.$$

§2.5.2 Product Notation

Products use the same indexing convention, with multiplication in place of addition. For $n \geq 1$,

$$\prod_{i=1}^n i = 1 \cdot 2 \cdot 3 \cdots n = n!.$$

§2.5.3 Sums and Products with One or No Terms

When the lower and upper bounds agree, the index takes a single value, so the sum or product consists of one term:

$$\sum_{i=m}^m x_i = x_m \quad \text{and} \quad \prod_{i=m}^m x_i = x_m.$$

When the upper bound is smaller than the lower bound, no integer lies in the range, and the sum or product is *empty*. By convention an empty sum is 0 and an empty product is 1. For example,

$$\sum_{j=5}^4 x_j = 0 \quad \text{and} \quad \prod_{i=10}^5 i^2 = 1.$$

Remark. These values are the identities for $+$ and $\cdot$: adding 0 or multiplying by 1 changes nothing. With this convention the splitting identity above also holds for $n = 1$, where the first piece is empty: $\sum_{i=1}^0 x_i + \sum_{i=1}^k x_i = 0 + \sum_{i=1}^k x_i$. It also gives $0! = \prod_{i=1}^0 i = 1$.

In practice

- **Reading a sum.** Identify the index, its first and last value, and the term as a function of the index; count the terms as $b - a + 1$ (zero when $b < a$).
- **Rewriting a sum.** To compare two sums, shift one index so both have the same bounds, or split off the first or last term.

§3 Logical Analysis of Mathematical Statements

§3.1 Truth Tables and Negation

Truth tables list all possible truth values of the component statements. Comparing complete columns lets us test logical equivalence.

Definition 3.1.1 (Negation)

The truth value for “not A ”, written symbolically as $\neg A$, is defined by the truth table given below.

A	$\neg A$
T	F
F	T

We also refer to $\neg A$ as the negation of A .

Here A is a **statement variable**: it stands for an arbitrary statement, so a truth table must list every possible truth value of A .

§3.1.1 Logical Equivalence

Definition 3.1.2 (Logical Equivalence)

Two compound statements are **logically equivalent**, written $\equiv$, if they have the same truth value in every row of their truth table, that is, for every assignment of truth values to their statement variables.

So a truth table can prove that two statements are logically equivalent.

Example 3.1.3

Given the truth table for A , $\neg A$, $\neg(\neg A)$,

A	$\neg A$	$\neg(\neg A)$
T	F	T
F	T	F

The first and last columns agree in every row, so this truth table serves as a proof that $A \equiv \neg(\neg A)$.

§3.2 Conjunction and Disjunction

Definition 3.2.1 (Compound Statement)

A compound statement is a statement composed of several individual statements, each of which is called a component statement.

Definition 3.2.2 (Conjunction)

The truth value for $A \wedge B$ is defined by the truth table

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

We also refer to $A \wedge B$ as the conjunction of A and B .

Each statement variable has two possible truth values, so with three statement variables we need $2^3 = 8$ rows to cover all the possibilities (and 2^n rows for n variables).

Example 3.2.3

The compound statement

$$(2 + 3 = 6) \wedge (\pi + 2 > 5)$$

is false, since the statement $2 + 3 = 6$ is false.

Definition 3.2.4 (Disjunction)

The truth value for A or B , written as $A \vee B$, is defined by the truth table

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

We also refer to $A \vee B$ as the **disjunction** of A and B .

Remark. The disjunction $A \vee B$ is the *inclusive* or: it is true whenever at least one of A , B is true, including the case where both are true. This differs from the *exclusive* or (xor), written $A \oplus B$, which is true when exactly one of A , B is true and false when both are true. Thus $A \vee B$ and $A \oplus B$ agree in every case except $A = B = \text{T}$, where $A \vee B$ is true but $A \oplus B$ is false.

Example 3.2.5

The compound statement

$$(2 + 3 = 6) \vee (\pi + 2 \geq 5)$$

is true, since its second component $\pi + 2 \geq 5$ is true.

§3.2.1 De Morgan's Laws**Example 3.2.6**

Use a truth table to prove that

$$\neg(A \vee B) \equiv (\neg A \wedge \neg B)$$

A	B	$A \vee B$	$\neg(A \vee B)$	$\neg A$	$\neg B$	$\neg A \wedge \neg B$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

The columns for $\neg(A \vee B)$ and $\neg A \wedge \neg B$ agree in all four rows.

A similar equivalence is

$$\neg(A \wedge B) \equiv ((\neg A) \vee (\neg B)).$$

Together, these two equivalences are called De Morgan's Laws: negating swaps $\wedge$ and $\vee$.

Theorem 3.2.7 (De Morgan's Laws)

For statement variables A and B , we have

$$\neg(A \vee B) \equiv (\neg A \wedge \neg B)$$

$$\neg(A \wedge B) \equiv (\neg A \vee \neg B)$$

§3.2.2 Laws of Conjunction and Disjunction

Proposition 3.2.8 (Logical Laws)

For statements A, B, C , the following rules hold for the logical operations $\wedge$ and $\vee$:

Commutative Laws

$$A \vee B \equiv B \vee A$$

$$A \wedge B \equiv B \wedge A$$

Associative Laws

$$A \vee (B \vee C) \equiv (A \vee B) \vee C$$

$$A \wedge (B \wedge C) \equiv (A \wedge B) \wedge C$$

Distributive Laws

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$$

In practice

- **Building a truth table.** For n statement variables, list all 2^n assignments. Compute inner expressions first, then compare the final columns in every row.
- **Negating a compound statement.** De Morgan's Laws swap $\wedge$ and $\vee$ while negating both components. Keep parentheses so that the scope of each negation is explicit.

§3.3 Implication

An implication specifies what must follow when its hypothesis is true. Its converse and contrapositive reverse the roles of the two statements in different ways.

Definition 3.3.1 (Implication)

The truth value for “ A implies B ”, written symbolically as $A \implies B$, is defined by the truth table

A	B	$A \implies B$
T	T	T
T	F	F
F	T	T
F	F	T

We also refer to $A \implies B$ as an implication.

In $A \implies B$, the statement A is the **hypothesis** (what we assume to be true) and B is the **conclusion** (what we must then show). The only way for an implication to be false is a true hypothesis with a false conclusion. When A is false, $A \implies B$ is true no matter what B is (it is *vacuously* true).

Example 3.3.2

$$(\forall x \in \mathbb{R}) \left[(x > 2) \implies (x^2 > 4) \right]$$

This is true. If $x > 2$, then $x > 0$, so $x^2 = x \cdot x > 2x > 4$. If $x \leq 2$, the hypothesis is false, so the implication is vacuously true for that x .

Remark. An implication with an “or” in its hypothesis splits into two implications:

$$((A \vee B) \implies C) \equiv (A \implies C) \wedge (B \implies C).$$

This is the logic behind a proof by cases.

§3.4 Converse and Contrapositive

§3.4.1 Converse

The implication $B \implies A$ is called the converse of $A \implies B$.

A	B	$A \implies B$	$B \implies A$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Remark. $(A \implies B) \not\equiv (B \implies A)$: the two columns differ in the second and third rows. For instance, $\forall x \in \mathbb{R}, (x > 2) \implies (x^2 > 4)$ is true, but its converse $\forall x \in \mathbb{R}, (x^2 > 4) \implies (x > 2)$ is false (take $x = -3$).

§3.4.2 Contrapositive

The implication $(\neg B) \implies (\neg A)$ is called the contrapositive of $A \implies B$.

A	B	$\neg A$	$\neg B$	$A \implies B$	$(\neg B) \implies (\neg A)$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Remark. $(A \implies B) \equiv ((\neg B) \implies (\neg A))$: an implication and its contrapositive always have the same truth value.

§3.4.3 Rewriting and Negating an Implication

An implication is also logically equivalent to a disjunction, as shown in the following truth table:

A	B	$\neg A$	$A \implies B$	$\neg A \vee B$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

That is, $(A \implies B) \equiv (\neg A \vee B)$.

Remark. The negation of an implication is not an implication. Negating $\neg A \vee B$ with De Morgan's Law gives

$$\neg(A \implies B) \equiv A \wedge \neg B,$$

which matches the only row where $A \implies B$ is false.

§3.4.4 If and Only If

Definition 3.4.1 (If and Only If)

The truth value for “ A if and only if B ”, written symbolically as $A \iff B$, is defined by the truth table

A	B	$A \iff B$
T	T	T
T	F	F
F	T	F
F	F	T

Thus $A \iff B$ is true exactly when A and B have the same truth value, and

$$(A \iff B) \equiv (A \implies B) \wedge (B \implies A).$$

To prove an “if and only if” statement, we prove an implication and its converse.

§4 Methods of Mathematical Proof

§4.1 Proving Universally Quantified Statements

A **proposition** is a mathematical statement established by proof. A **theorem** is a particularly significant proposition. A **lemma** is a subsidiary proposition. A **corollary** is a proposition that follows almost immediately from a theorem.

Example 4.1.1

The law of cosines is a theorem: in a triangle with sides a, b, c , where θ is the angle between a and b ,

$$c^2 = a^2 + b^2 - 2ab \cos \theta.$$

The Pythagorean theorem is a corollary: taking $\theta = \pi/2$ gives $\cos \theta = 0$, so $c^2 = a^2 + b^2$.

§4.1.1 An Arbitrary Element

To prove $\forall x \in S, P(x)$, let x be an *arbitrary* element of S , using only the fact that $x \in S$, and show that $P(x)$ is true. Since nothing special about x was assumed, the argument works for every element of S .

The following inequalities are proved by rewriting a difference as a square.

Proposition 4.1.2

$$\forall x \in \mathbb{R}, x^2 + 1 \geq 2x.$$

Remark (Proof idea). Moving everything to one side gives $x^2 - 2x + 1 \geq 0$, and the left side is the perfect square $(x - 1)^2$. Since squares of real numbers are never negative, we start from $(x - 1)^2 \geq 0$ and work forward.

Proof. Let x be an arbitrary real number. Therefore, $x - 1$ must also be a real number, and hence

$$(x - 1)^2 \geq 0.$$

Expanding the left side, we get

$$x^2 - 2x + 1 \geq 0.$$

Adding $2x$ to both sides yields

$$x^2 + 1 \geq 2x.$$

Q.E.D.

Proposition 4.1.3

$$\forall x, y \in \mathbb{R}, x^4 + x^2y + y^2 \geq 5x^2y - 3y^2.$$

Remark (Proof idea). Moving everything to the left side gives $x^4 - 4x^2y + 4y^2 \geq 0$, and $x^4 - 4x^2y + 4y^2 = (x^2 - 2y)^2$. This explains the choice of $x^2 - 2y$ below.

Proof. Let x and y be arbitrary real numbers. Then $x^2 - 2y$ is also a real number. So

$$(x^2 - 2y)^2 \geq 0.$$

Expanding, we get

$$x^4 - 4x^2y + 4y^2 \geq 0.$$

Adding $5x^2y$ to both sides, we have

$$x^4 + x^2y + 4y^2 \geq 5x^2y.$$

Now, subtract $3y^2$ from both sides:

$$x^4 + x^2y + y^2 \geq 5x^2y - 3y^2.$$

Q.E.D.

§4.1.2 Proof by Cases

Proposition 4.1.4

$$\forall x, y \in \mathbb{R}, \max(x, y) = \frac{x + y + |x - y|}{2}.$$

Remark (Proof idea). Both $\max(x, y)$ and $|x - y|$ are defined by cases depending on which of x, y is larger, so we split into the cases $x \geq y$ and $x < y$. These cases cover every pair of real numbers.

Proof. Let x and y be arbitrary real numbers.

Case 1: $x \geq y$. Then $\max(x, y) = x$ and $x - y \geq 0$, so $|x - y| = x - y$.

Then

$$\frac{x + y + |x - y|}{2} = \frac{x + y + x - y}{2} = \frac{2x}{2} = x = \max(x, y).$$

So the claim holds when $x \geq y$.

Case 2: $x < y$. Then $\max(x, y) = y$ and $x - y < 0$, so

$$|x - y| = -(x - y) = y - x.$$

Then

$$\frac{x + y + |x - y|}{2} = \frac{x + y + y - x}{2} = \frac{2y}{2} = y = \max(x, y).$$

The claim holds in both cases, so it holds for all $x, y \in \mathbb{R}$.

Q.E.D.

§4.1.3 Disproving a Universal Statement

Since $\neg(\forall x \in S, P(x)) \equiv \exists x \in S, \neg P(x)$, to disprove a universal statement it suffices to give one **counterexample**: a specific $x \in S$ for which $P(x)$ is false.

Example 4.1.5

Disprove:

$$\forall x \in \mathbb{R}, (x^2 - 1)^2 > 0.$$

Solution. Take $x = 1$. Then

$$(x^2 - 1)^2 = (1 - 1)^2 = 0 \not> 0.$$

So $x = 1$ is a counterexample, and the statement is false.

In practice

- **Universal proof.** Start with an arbitrary element of the stated domain. Use the hypotheses to derive the claim without choosing a special value of that element.

- **Cases.** Choose cases that cover the entire domain and prove the same conclusion in each. For absolute values, the sign of the expression inside determines the cases.
- **Counterexample.** Give one element in the domain and verify that it fails the claimed condition.

Exercises

- (i) Prove $\forall \theta \in \mathbb{R}, \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$.
- (ii) Prove $\forall x \in \mathbb{R}, |x - 3| + 2|x + 2| \geq 5$.

§4.2 Contrapositive and Elimination Methods

Logical equivalences let us replace a claim with a form whose hypotheses are easier to use. Several examples below use divisibility.

Definition 4.2.1 (Divisibility)

Let $a, b \in \mathbb{Z}$. We say a **divides** b , written $a \mid b$, if

$$\exists k \in \mathbb{Z}, b = ak.$$

Otherwise we write $a \nmid b$.

§4.2.1 Proof by Contrapositive

Definition 4.2.2 (Proof by Contrapositive)

Since $(A \implies B) \equiv (\neg B \implies \neg A)$, we can prove $(A \implies B)$ by proving that $((\neg B) \implies (\neg A))$. This method is useful when a direct proof of $A \implies B$ is awkward, but assuming $\neg B$ gives a concrete hypothesis that is easier to work with.

Proposition 4.2.3

For every integer x , if $4 \nmid x$ then $36 \nmid 3x$.

Remark (Proof idea). The hypothesis $4 \nmid x$ says only what x is *not*, which is hard to use directly, while $36 \mid 3x$ gives an equation involving an integer multiple.

Proof. We prove the contrapositive

$$(36 \mid 3x) \implies (4 \mid x).$$

Let x be an arbitrary integer. Assume that $36 \mid 3x$. Thus, there exists an integer k such that $3x = 36k$. Dividing both sides by 3,

$$x = 12k = 4 \cdot (3k) \text{ and } 3k \in \mathbb{Z},$$

so $4 \mid x$.

Therefore, by the contrapositive method,

$$(\forall x \in \mathbb{Z}) [4 \nmid x \implies 36 \nmid 3x].$$

Q.E.D.

Proposition 4.2.4

$$(\forall a, b \in \mathbb{R}) [(ab \in \mathbb{R} \setminus \mathbb{Q}) \implies (a \in \mathbb{R} \setminus \mathbb{Q} \vee b \in \mathbb{R} \setminus \mathbb{Q})]$$

Remark (Proof idea). Irrationality is a “not” condition, while rationality gives us fractions to compute with. Negating the disjunction in the conclusion makes both factors rational.

Proof. By De Morgan's Law, the negation of the conclusion is " $a \in \mathbb{Q}$ and $b \in \mathbb{Q}$ ". Thus we prove the contrapositive: $\forall a, b \in \mathbb{R}$, if $a \in \mathbb{Q}$ and $b \in \mathbb{Q}$, then $ab \in \mathbb{Q}$.

Let a, b be arbitrary in $\mathbb{R}$. Assume that a, b are both rational.

Then there exist $p, q, r, s \in \mathbb{Z}$ with $q \neq 0$ and $s \neq 0$ such that

$$a = \frac{p}{q} \quad \text{and} \quad b = \frac{r}{s}.$$

Thus,

$$a \cdot b = \frac{pr}{qs} \quad \text{and} \quad qs \neq 0,$$

where $pr, qs \in \mathbb{Z}$. Hence $ab \in \mathbb{Q}$.

By the contrapositive,

$$\forall a, b \in \mathbb{R}, (ab \notin \mathbb{Q}) \implies (a \notin \mathbb{Q} \vee b \notin \mathbb{Q}).$$

Q.E.D.

§4.2.2 Elimination Method

Definition 4.2.5 (Elimination Method)

Let A, B, C be statement variables.

$$(A \implies (B \vee C)) \equiv ((A \wedge \neg B) \implies C).$$

This is useful when the conclusion is a disjunction $B \vee C$: instead of proving the disjunction directly, we assume one disjunct fails ($\neg B$) and show the other (C) must then hold.

Since $B \vee C \equiv C \vee B$, we may instead eliminate C and prove

$$(A \wedge \neg C) \implies B.$$

Proposition 4.2.6

For all $x \in \mathbb{R}$, if $|2x - 6| = 4$, then $x > 3$ or $x = 1$.

Proof. The negation of $x > 3$ is $x \leq 3$, so the elimination form is

$$\forall x \in \mathbb{R}, [(|2x - 6| = 4) \wedge (x \leq 3)] \implies x = 1.$$

Let x be an arbitrary real number. Assume $|2x - 6| = 4$ and $x \leq 3$, and prove $x = 1$.

Since $|2x - 6| = 4$, we have

$$2x - 6 = \pm 4.$$

So $2x = 10$ or $2x = 2$, meaning

$$x = 5 \quad \text{or} \quad x = 1.$$

Since $x \leq 3$, the value $x = 5$ is ruled out, so

$$x = 1.$$

Q.E.D.

§4.2.3 Using a Universal Hypothesis

Proposition 4.2.7

$$\forall a, b, c \in \mathbb{Z}, (\forall x \in \mathbb{Z}, a \mid (bx + c)) \implies (a \mid (c - 2b)).$$

Remark (Proof idea). The hypothesis holds for *every* integer x , so we may instantiate it at any x . Choose $x = -2$ to make $bx + c$ equal the expression in the conclusion.

Proof. Let a, b, c be arbitrary integers. Assume for all $x \in \mathbb{Z}$, $a \mid (bx + c)$.

Since this holds for every integer x , it holds in particular when $x = -2$, so we get

$$a \mid (b(-2) + c),$$

that is

$$a \mid (c - 2b).$$

Q.E.D.

§4.2.4 Integer Combinations

Proposition 4.2.8 (Divisibility of Integer Combinations (DIC))

Let $a, b, c \in \mathbb{Z}$. If $a \mid b$ and $a \mid c$, then for all $x, y \in \mathbb{Z}$,

$$a \mid (bx + cy).$$

Remark (Proof idea). Divisibility means b and c are each an integer multiple of a . Any integer combination $bx + cy$ is then a sum of multiples of a , hence itself a multiple of a — this is exactly what licenses factoring a out below.

Proof. Let $a, b, c \in \mathbb{Z}$. Assume $a \mid b$ and $a \mid c$. Then there exist integers k_1, k_2 such that

$$b = ak_1 \quad \text{and} \quad c = ak_2.$$

Let $x, y \in \mathbb{Z}$ be arbitrary. Then

$$bx + cy = ak_1x + ak_2y = a(k_1x + k_2y).$$

Since $k_1x + k_2y \in \mathbb{Z}$, this shows $a \mid (bx + cy)$.

Q.E.D.

Proposition 4.2.9 (Converse of DIC)

Let $a, b, c \in \mathbb{Z}$. If for all $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$, then $a \mid b$ and $a \mid c$.

Remark (Proof idea). The hypothesis holds for *every* choice of x, y , so we are free to pick whichever values isolate b and c on their own: $(x, y) = (1, 0)$ leaves $bx + cy = b$, and $(x, y) = (0, 1)$ leaves $bx + cy = c$.

Proof. Let a, b, c be arbitrary integers. Assume that for all $x, y \in \mathbb{Z}$, $a \mid (bx + cy)$.

Taking $x = 1, y = 0$, we get

$$a \mid (b \cdot 1 + c \cdot 0), \quad \text{that is,} \quad a \mid b.$$

Taking $x = 0, y = 1$, we get

$$a \mid (b \cdot 0 + c \cdot 1), \quad \text{that is,} \quad a \mid c.$$

Therefore $a \mid b$ and $a \mid c$.

Q.E.D.

In practice

- **Contrapositive.** To prove $A \implies B$, assume $\neg B$ and derive $\neg A$. Negate compound statements using De Morgan's Laws.
- **Elimination.** To prove $A \implies (B \vee C)$, assume A and $\neg B$, then prove C ; alternatively, assume A and $\neg C$ and prove B .
- **Universal hypotheses.** A hypothesis that holds for every value may be used at a convenient value. Choose values that isolate the desired expression, as $x = -2$ or $(x, y) = (1, 0), (0, 1)$ do above.

§4.3 Proof by Contradiction

Definition 4.3.1 (Contradiction)

Given a statement A , the logical expression $A \wedge (\neg A)$ is called a contradiction.

A contradiction is false in every row of its truth table. To prove a statement S by contradiction, assume $\neg S$ and derive a contradiction; since a true assumption cannot lead to a false statement, $\neg S$ must be false, so S is true. For an implication $A \implies B$, we assume A together with $\neg B$.

Proposition 4.3.2

$\forall n \in \mathbb{Z}$, if n^2 is even, then n is even.

Remark (Proof idea). If the conclusion fails, n is odd and can be written as $2k + 1$. Squaring this expression makes n^2 odd as well, contradicting the hypothesis.

Proof. Let n be an arbitrary integer and assume n^2 is even.

Suppose, for contradiction, that n is odd. Then there is some $k \in \mathbb{Z}$ such that

$$n = 2k + 1.$$

Squaring gives

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

and $2k^2 + 2k \in \mathbb{Z}$ (since $k \in \mathbb{Z}$), so n^2 is odd.

This contradicts the assumption that n^2 is even.

Therefore n cannot be odd, so n is even.

Q.E.D.

Remark. The contradiction here is $(n^2 \text{ is even}) \wedge (n^2 \text{ is odd})$, an instance of $A \wedge (\neg A)$: no integer is both even and odd.

In practice

- **Contradiction template.** Write “Suppose, for contradiction, that $\neg S$ ” (for $A \implies B$: assume A and $\neg B$), derive consequences until some statement and its negation both hold, then conclude S .
- **When to use it.** Contradiction helps when $\neg S$ gives something concrete to work with, as “ n is odd” gives $n = 2k + 1$ while “ n^2 is even” says nothing directly about n .

Exercises

- Prove there is no greatest integer.
- Prove $\forall a, b \in \mathbb{Z}, a \geq 2 \implies (a \nmid b) \vee (a \nmid (b + 1))$.
- Prove $\sqrt{2}$ is irrational.

§4.4 Proving If and Only If Statements

The statement $A \iff B$ means $(A \implies B) \wedge (B \implies A)$. Prove both directions to establish the equivalence.

Example 4.4.1

Prove that for all non-negative $x, y \in \mathbb{R}$,

$$\frac{x + y}{2} = \sqrt{xy} \iff x = y.$$

Remark (Proof idea). If $x = y$, evaluate both sides. In the other direction, square the equality and rearrange it into $(x - y)^2 = 0$.

Proof. Let $x, y \in \mathbb{R}$ be arbitrary and non-negative.

($\Leftarrow$) Suppose $x = y$. Then

$$\frac{x+y}{2} = y = \sqrt{y^2} = \sqrt{xy},$$

where $\sqrt{y^2} = y$ because $y \geq 0$.

($\Rightarrow$) Suppose $\frac{x+y}{2} = \sqrt{xy}$. Squaring both sides and multiplying by 4 gives

$$x^2 + 2xy + y^2 = 4xy,$$

so $x^2 - 2xy + y^2 = (x - y)^2 = 0$. Hence $x - y = 0$, that is, $x = y$.

Q.E.D.

In practice

- **Two directions.** Label the parts ($\Rightarrow$) and ($\Leftarrow$); in each, assume one side and derive the other. Proving only one direction proves an implication, not the equivalence.

§4.5 Proving Uniqueness

To prove that there is a unique $x \in S$ satisfying $P(x)$, show both that such an x exists and that any two elements satisfying P are equal. Proving only the second part does not establish existence.

Proposition 4.5.1

For all $a, b \in \mathbb{Z}$, if $a \neq 0$ and $a \mid b$, then there is a unique integer k such that $b = ka$.

Remark (Proof idea). Divisibility gives existence directly. For uniqueness, compare two possible multiples of the same nonzero integer a .

Proof. Let $a, b \in \mathbb{Z}$ be arbitrary. Assume $a \neq 0$ and $a \mid b$.

Existence. By the definition of divisibility, there is some $k \in \mathbb{Z}$ such that $b = ka$.

Uniqueness. Suppose $k, m \in \mathbb{Z}$ both satisfy the equation. Then $ka = b = ma$. Since $a \neq 0$, dividing by a gives $k = m$. *Q.E.D.*

Remark. The condition $a \neq 0$ is needed for uniqueness. If $a = b = 0$, then $0 \mid 0$, but every integer k satisfies $0 = k \cdot 0$.

In practice

- **Existence, then uniqueness.** First produce (or cite) one object with the property. Then assume two objects x and y both have it and show $x = y$. Alternatively, assume two *different* objects have it and derive a contradiction.

Exercises

- Use a proof by contradiction to show that two non-parallel lines have a unique intersection point.

§5 Sets

§5.1 Basic Set Notation and Operations

Set notation describes which objects belong to a collection. Union, intersection, and difference combine sets by applying logical conditions to membership.

Definition 5.1.1 (Empty Set and Cardinality)

A *set* is a collection of distinct elements. The *empty set* $\emptyset$ has no elements. For a finite set S , its *cardinality* $|S|$ is its number of elements.

We can describe a set by listing its elements, selecting elements with a property, or generating values:

- Listing: $\{2, 3, 4, 6\}$.
- Selection: $\{n \in \mathbb{Z} : 0 < n < 10 \text{ and } n \text{ is even}\} = \{2, 4, 6, 8\}$.
- Generation: $\{2k : k \in \{1, 2, 3, 4\}\} = \{2, 4, 6, 8\}$.

Example 5.1.2

Order and repetition do not change a set: $\{1, 2, 2\} = \{2, 1\}$. Also, $|\emptyset| = 0$, while $|\{\emptyset\}| = 1$: the latter set contains the empty set as its one element.

§5.1.1 Set-Builder Notation

Definition 5.1.3 (Set-Builder Notation)

The *selection form* $\{x \in U : P(x)\}$ contains exactly the elements of U for which $P(x)$ is true. The *generation form* $\{f(x) : x \in U\}$ contains the values $f(x)$ for $x \in U$; its elements are these outputs, not the inputs x .

Example 5.1.4

$\{x^2 : x \in \{-2, -1, 0, 1, 2\}\} = \{0, 1, 4\}$: different inputs may give the same output, which appears only once.

We can combine the forms: $\{f(x) : x \in U, P(x)\}$ first selects the inputs satisfying P , then forms their outputs. For example,

$$\{3a : a \in \mathbb{Z}, 2 \mid a\} = \{6k : k \in \mathbb{Z}\}.$$

§5.1.2 Intervals

For real numbers $a < b$, square brackets include an endpoint and round brackets exclude it:

$$\begin{aligned} [a, b] &= \{x \in \mathbb{R} : a \leq x \leq b\}, & (a, b) &= \{x \in \mathbb{R} : a < x < b\}, \\ [a, b) &= \{x \in \mathbb{R} : a \leq x < b\}, & (a, b] &= \{x \in \mathbb{R} : a < x \leq b\}. \end{aligned}$$

§5.1.3 Set Operations

Definition 5.1.5 (Set Operations)

The *union* $S \cup T$ collects elements in either set; the *intersection* $S \cap T$ keeps elements in both. The *difference* $S - T$ keeps elements in S but not T :

$$\begin{aligned} S \cup T &= \{x : x \in S \text{ or } x \in T\}, & S \cap T &= \{x : x \in S \text{ and } x \in T\}, \\ S - T &= \{x : x \in S \text{ and } x \notin T\}. \end{aligned}$$

When S is a subset of a universal set U , its *complement* is $\bar{S} = U - S$.

Example 5.1.6

Let $A = [-1, 2)$ and $B = (0, 4]$. Then

$$\begin{aligned} A \cup B &= [-1, 4], & A \cap B &= (0, 2), \\ A - B &= [-1, 0], & B - A &= [2, 4]. \end{aligned}$$

The point 2 belongs to B but not A , so it is excluded from the intersection and included in $B - A$.

For a set S contained in the universal set U , the following identities hold, with complements taken relative to U :

$$\begin{aligned} S \cup \emptyset &= S, & S \cap \emptyset &= \emptyset, \\ S \cup U &= U, & S \cap U &= S, \\ S \cup S &= S, & S \cap S &= S, \\ S \cup \bar{S} &= U, & S \cap \bar{S} &= \emptyset. \end{aligned}$$

In practice

- **Reading set-builder notation.** Before the colon: what the elements look like. After it: where the variable ranges and which conditions it must satisfy. To test membership of y in $\{f(x) : x \in U\}$, look for some $x \in U$ with $f(x) = y$.
- **Operations on intervals.** Draw both intervals on a number line and check each endpoint separately: it belongs to the result exactly when the defining condition (or, and, and not) holds for it.

§5.2 Subsets and Set Equality

To compare two sets we ask whether every element of one belongs to the other. This gives the subset relation, and two subset relations together give set equality.

§5.2.1 Disjoint Sets and Subsets**Definition 5.2.1 (Disjoint)**

Two sets S and T are *disjoint* when they have no element in common, that is,

$$S \cap T = \emptyset.$$

Definition 5.2.2 (Subset)

A set S is a *subset* of a set T , written $S \subseteq T$, when every element of S belongs to T .

Definition 5.2.3 (Proper Subset)

A set S is a *proper subset* of a set T , written $S \subsetneq T$, when $S \subseteq T$ and some element of T does not belong to S .

Remark. If $S \subseteq T$, we also say T is a *superset* of S . If $S \subsetneq T$, we say T is a *proper superset* of S . We write $S \not\subseteq T$ when S is not a subset of T , that is, when some element of S does not belong to T .

Example 5.2.4 – $S \subseteq S$ for every set S .

- $\emptyset \subseteq S$ for every set S : the empty set has no element that could fail to belong to S .
- $\mathbb{N} \subseteq \mathbb{Z}$; in fact $\mathbb{N} \subsetneq \mathbb{Z}$, since $0 \in \mathbb{Z}$ but $0 \notin \mathbb{N}$.
- $\mathbb{Z} \not\subseteq \mathbb{N}$, since $-1 \in \mathbb{Z}$ but $-1 \notin \mathbb{N}$.

§5.2.2 Proving a Subset Relation

To prove that $S \subseteq T$, prove the universally quantified implication

$$\forall x \in \mathcal{U}, (x \in S) \implies (x \in T),$$

where $\mathcal{U}$ is the universe of discourse: let x be an arbitrary element of S and show that $x \in T$. To prove $S \not\subseteq T$, one counterexample suffices: an element of S that is not in T .

Example 5.2.5

Let $A = \{n \in \mathbb{N} : 4 \mid (n - 3)\}$ and $B = \{2k + 1 : k \in \mathbb{Z}\}$.

- (i) Prove that $A \subseteq B$.
- (ii) Prove that $A \subsetneq B$.

Remark (Proof idea). Elements of B are the odd integers. For (i), unpack $4 \mid (n - 3)$ into $n = 4k + 3$ and rewrite it in the form $2(\cdot) + 1$. For (ii), find an odd integer that is not in A .

Proof. (i) Let $n \in \mathbb{N}$ and assume $n \in A$. By the definition of A , $4 \mid (n - 3)$, so there is some $k \in \mathbb{Z}$ such that $n - 3 = 4k$. Then

$$n = 4k + 3 = 4k + 2 + 1 = 2(2k + 1) + 1.$$

Since $2k + 1 \in \mathbb{Z}$, we have $n \in B$. Therefore $A \subseteq B$.

(ii) By (i), $A \subseteq B$. Now $1 = 2 \cdot 0 + 1 \in B$, but $1 - 3 = -2$ and $4 \nmid -2$, so $1 \notin A$. Hence B has an element that is not in A , and $A \subsetneq B$. Q.E.D.

§5.2.3 Proving Set Equality

Two sets are equal when they have exactly the same elements. So, to prove $S = T$, we prove both

$$S \subseteq T \quad \text{and} \quad T \subseteq S.$$

Example 5.2.6

Let $S = \{-1, 0, 1\}$ and $T = \{x \in \mathbb{R} : x^3 = x\}$. Prove that $S = T$.

Proof. $S \subseteq T$. Let $x \in S$. Then x is -1 , 0 , or 1 , and we check each case:

$$(-1)^3 = -1, \quad 0^3 = 0, \quad 1^3 = 1.$$

In each case $x^3 = x$, so $x \in T$.

$T \subseteq S$. Let $x \in T$. Then $x^3 = x$, so

$$0 = x^3 - x = x(x^2 - 1) = x(x - 1)(x + 1).$$

A product of real numbers is 0 only if one factor is 0, so $x = 0$, $x = 1$, or $x = -1$. Thus $x \in S$.

Therefore $S = T$. Q.E.D.

Example 5.2.7

Prove that for all sets S and T , $S = T$ if and only if $S \cap T = S \cup T$.

Remark (Proof idea). Each direction is a set equality, proved by two subset relations. For $(\Leftarrow)$, an element of S lies in $S \cup T$, hence in $S \cap T$, hence in T ; the same chain works with S and T swapped.

Proof. $(\Rightarrow)$ Suppose $S = T$. We show $S \cap T = S \cup T$.

First, $S \cap T \subseteq S \cup T$: if $x \in S \cap T$, then $x \in S$ and $x \in T$; in particular $x \in S$, so $x \in S \cup T$.

Next, $S \cup T \subseteq S \cap T$: if $x \in S \cup T$, then $x \in S$ or $x \in T$. Since $S = T$, in either case $x \in S$ and $x \in T$, so $x \in S \cap T$.

(Alternatively, substitute $T = S$: $S \cap T = S \cap S = S = S \cup S = S \cup T$.)

($\Leftarrow$) Suppose $S \cap T = S \cup T$. We show $S = T$.

First, $S \subseteq T$: for $x \in S$,

$$x \in S \implies x \in S \cup T \implies x \in S \cap T \implies x \in T.$$

Second, $T \subseteq S$: for $x \in T$,

$$x \in T \implies x \in S \cup T \implies x \in S \cap T \implies x \in S.$$

The middle step in each chain uses the hypothesis $S \cup T = S \cap T$.

Therefore $S = T$.

Q.E.D.

In practice

- **Element chasing.** To prove $S \subseteq T$, write “Let $x \in S$ ”, unpack the definition of S , and rebuild the definition of T . To disprove it, exhibit one element of S not in T .
- **Set equality.** Prove $S \subseteq T$ and $T \subseteq S$ separately. For a finite set such as $\{-1, 0, 1\}$, the first inclusion is a finite case check.
- **Proper subset.** Prove $S \subseteq T$, then exhibit one element of T outside S .

§6 Mathematical Induction

§6.1 Principle of Mathematical Induction

Induction proves a statement for every natural number at once: we check the first case, then show that each case implies the next.

Axiom (The Principle of Mathematical Induction)

Let $P(n)$ be an open sentence that depends on $n \in \mathbb{N}$. If

- (1) $P(1)$ is true, and
- (2) $\forall k \in \mathbb{N}, P(k) \implies P(k+1)$,

then $P(n)$ is true for all $n \in \mathbb{N}$.

Condition (2) is a universally quantified implication: it does not assert $P(k)$, only that $P(k)$ would force $P(k+1)$. Together with (1), it yields $P(1) \implies P(2) \implies P(3) \implies \dots$.

§6.1.1 Proof by Mathematical Induction

To prove a universally quantified statement $\forall n \in \mathbb{N}, P(n)$:

- **Base case.** Prove $P(1)$.
- **Inductive step.** Let $k \in \mathbb{N}$ be arbitrary, assume $P(k)$ (the *inductive hypothesis*), and prove $P(k+1)$.

Example 6.1.1

Prove that for all $n \in \mathbb{N}$,

$$\sum_{i=1}^n i(i+1) = \frac{1}{3} n(n+1)(n+2).$$

Remark (Proof idea). The sum up to $k+1$ is the sum up to k plus one new term. Replace the first part using the inductive hypothesis, then factor out $(k+1)(k+2)$.

Proof. Let $P(n)$ be the statement above.

Base case. $P(1)$ is true, since

$$\sum_{i=1}^1 i(i+1) = 1 \cdot 2 = 2 = \frac{1}{3} \cdot 1 \cdot 2 \cdot 3.$$

Inductive step. Let k be an arbitrary natural number and assume $P(k)$, that is,

$$\sum_{i=1}^k i(i+1) = \frac{1}{3} k(k+1)(k+2).$$

We prove $P(k+1)$. Splitting off the last term and using the inductive hypothesis,

$$\begin{aligned} \sum_{i=1}^{k+1} i(i+1) &= \sum_{i=1}^k i(i+1) + (k+1)(k+2) \\ &= \frac{1}{3} k(k+1)(k+2) + (k+1)(k+2) \\ &= (k+1)(k+2) \left(\frac{k}{3} + 1 \right) \\ &= \frac{1}{3} (k+1)(k+2)(k+3). \end{aligned}$$

This is $P(k+1)$. By the Principle of Mathematical Induction, $P(n)$ is true for all $n \in \mathbb{N}$.

Q.E.D.

§6.1.2 Starting at a Later Base Case

If a statement is claimed only for integers $n \geq b$, the base case is $n = b$ and the inductive step assumes $k \geq b$. The argument is the same: the base case starts the chain at b , and each step moves it one further.

Example 6.1.2

Prove that $n! > 2^n$ for every integer $n \geq 4$.

Remark (Proof idea). Going from k to $k + 1$ multiplies the left side by $k + 1$ and the right side by 2. Since $k + 1 \geq 2$, the left side grows at least as fast.

Proof. We use induction on n , starting at $n = 4$.

Base case. For $n = 4$, the left side is $4! = 24$ and the right side is $2^4 = 16$. Since $24 > 16$, the statement holds for $n = 4$.

Inductive step. Let k be an arbitrary integer with $k \geq 4$, and assume $k! > 2^k$. We prove $(k + 1)! > 2^{k+1}$. Multiplying the inductive hypothesis by $k + 1 > 0$ gives

$$(k + 1)! = (k + 1) k! > (k + 1) 2^k.$$

Since $k \geq 4$, we have $k + 1 \geq 5 \geq 2$, so

$$(k + 1)! > (k + 1) 2^k \geq 2 \cdot 2^k = 2^{k+1}.$$

Hence the statement holds for $k + 1$. By induction, $n! > 2^n$ for every integer $n \geq 4$.

Q.E.D.

Remark. The base case cannot be moved earlier: $3! = 6 < 8 = 2^3$. The inductive step alone proves nothing; it needs a true starting case.

In practice

- **Induction template.** State $P(n)$; prove the base case; write “Let k be arbitrary and assume $P(k)$ ”; write out $P(k + 1)$ as the goal; conclude “by the Principle of Mathematical Induction”.
- **Sums.** Split off the last term, $\sum_{i=1}^{k+1} t_i = \sum_{i=1}^k t_i + t_{k+1}$, so the inductive hypothesis applies to the first part.
- **Inequalities.** Compare the factor each side gains from k to $k + 1$, and use the range of k (here $k \geq 4$) to compare them.

§6.2 Principle of Strong Induction

Some inductive steps need more than the immediately preceding case. Strong induction makes all earlier cases available, which is useful for recurrences that depend on two or more previous terms.

§6.2.1 Why Use a Stronger Inductive Hypothesis?**Example 6.2.1**

A sequence $\{x_n\}$ is defined recursively by

$$x_1 = 4, \quad x_2 = 68, \quad x_n = 2x_{n-1} + 15x_{n-2} \quad \text{for all } n \geq 3.$$

Prove the closed form

$$x_n = 2(-3)^n + 10 \cdot 5^{n-1} \quad \text{for all } n \in \mathbb{N}.$$

Let $P(n)$ be the claimed formula. The first case holds because $2(-3) + 10 = 4 = x_1$. However, if we assume only $P(k)$, the recurrence

$$x_{k+1} = 2x_k + 15x_{k-1} \quad (k \geq 2)$$

still involves x_{k-1} , whose closed form is not supplied by that hypothesis. We need both $P(k)$ and $P(k - 1)$, as well as a second base case. This does not mean ordinary induction is invalid; it means this choice of inductive hypothesis does not provide everything the step needs.

§6.2.2 The Strong Induction Principle

Definition 6.2.2 (Principle of Strong Induction)

Let $P(n)$ be an open sentence that depends on $n \in \mathbb{N}$. If

- (1) $P(1)$ is true, and
- (2) for every $k \in \mathbb{N}$, if $P(1), P(2), \dots, P(k)$ are all true, then $P(k+1)$ is true,

then $P(n)$ is true for all $n \in \mathbb{N}$.

The step assumes every case up to k , not just the case k . It may use whichever of these earlier cases the argument needs.

To start at an integer b and use several base cases, fix an integer $B \geq b$. Prove $P(b), P(b+1), \dots, P(B)$. Then, for arbitrary $k \geq B$, assume $P(i)$ for every integer $b \leq i \leq k$ and prove $P(k+1)$. Here B is fixed by the proof; we do not prove base cases for every B .

§6.2.3 Returning to the Recurrence

Remark (Proof idea). Check the two initial terms, then substitute the formulas for both x_k and x_{k-1} . Group the powers of -3 and of 5 separately.

Proof. We prove $P(n)$ by strong induction on n .

Base cases. Both $P(1)$ and $P(2)$ hold, since

$$\begin{aligned} 2(-3)^1 + 10 \cdot 5^0 &= -6 + 10 = 4 = x_1, \\ 2(-3)^2 + 10 \cdot 5^1 &= 18 + 50 = 68 = x_2. \end{aligned}$$

Inductive step. Let $k \geq 2$ be an arbitrary integer. Assume $P(i)$ is true for every integer $1 \leq i \leq k$, that is,

$$x_i = 2(-3)^i + 10 \cdot 5^{i-1} \quad \text{for } 1 \leq i \leq k.$$

In particular, the hypothesis applies to both k and $k-1$. Using the recurrence and substituting these two cases,

$$\begin{aligned} x_{k+1} &= 2x_k + 15x_{k-1} \\ &= 2[2(-3)^k + 10 \cdot 5^{k-1}] + 15[2(-3)^{k-1} + 10 \cdot 5^{k-2}] \\ &= 4(-3)^k + 30(-3)^{k-1} + 20 \cdot 5^{k-1} + 150 \cdot 5^{k-2} \\ &= (-3)^{k-1}[4(-3) + 30] + 5^{k-2}[20 \cdot 5 + 150] \\ &= 18(-3)^{k-1} + 250 \cdot 5^{k-2} \\ &= 2(-3)^{k+1} + 10 \cdot 5^k. \end{aligned}$$

This is $P(k+1)$. By strong induction, the formula holds for every $n \in \mathbb{N}$.

Q.E.D.

In practice

- **Choose the hypothesis from the recurrence.** If the next term uses two preceding terms, check both initial cases and make both preceding formulas available in the inductive step.
- **Strong induction template.** State $P(n)$ and check the required base cases. For arbitrary k beyond them, assume every earlier case in the claimed range, prove $P(k+1)$, and conclude by strong induction.

Exercises

- (i) (Lecture.) Suppose $x_1 = 3$, $x_2 = 5$, and $x_n = 3x_{n-1} + 2x_{n-2}$ for all $n \geq 3$. Prove that $x_n < 4^n$ for every $n \in \mathbb{N}$.
- (ii) (Lecture.) An $m \times n$ rectangular chocolate bar consists of $N = mn$ unit squares. Prove that $N - 1$ breaks are required to separate the bar into unit squares.